

The Representation-Theoretic Uniqueness of (2,3): Why Physics is Always $SU(3) \times SU(2) \times U(1)$

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Abstract

We prove two theorems requiring no physical input. First, for any integer $N \geq 5$, the unique decomposition of $\mathbb{C}^N$ into atomic blocks that supports chiral fermions, CP violation, and connected spacetime is $(n_S, n_T) = (3, 2)$, yielding the gauge group $SU(3) \times SU(2) \times U(1)$. This holds regardless of N : repeated atomic blocks are rendered physically trivial by swap symmetry, and only one copy each of the atoms $\{2, 3\}$ survives. Second, the coupling constant of this unique structure, $\alpha_{GUT} = 6/(25\pi^2)$, is derived independently from three branches of mathematics — simplex geometry (Binet-Cauchy decomposition), random matrix theory (GUE sine kernel), and number theory (coprimality probability) — all originating from the algebraic uniqueness of $\mathbb{C}$ among division algebras. Numerical simulations confirm that swap annihilation does not create a spectral gap at rank 5; the distinction between chiral and trivial dimensions is purely representation-theoretic and invisible to spectroscopy. The Standard Model gauge group and its coupling constant are theorems of mathematics, not properties of a particular universe.

1. Introduction

The Standard Model of particle physics contains 26 free parameters and is built on the gauge group $SU(3) \times SU(2) \times U(1)$. The question “why this group?” has motivated decades of research in grand unification, string theory, and mathematical physics.

The DRLT framework (Discrete Relational Lattice Theory) proposes that starting from a single axiom — “points exist with pairwise relations” — the substrate $\mathbb{C}$, the dimension $d = 5$, and all Standard Model parameters can be derived. The present work strengthens the dimensional argument: we show that the conclusion does not depend on $d = 5$ at all. For *any* dimension N , the unique physically non-trivial content is $(2, 3)$, and its coupling constant is the mathematical constant $6/(25\pi^2)$.

The paper is organized as follows. Section 2 reviews the atomic decomposition and swap annihilation. Section 3 presents numerical simulations revealing that swap annihilation creates no spectral gap at rank 5. Section 4 analyzes the eigenvalue structure in detail, identifying a soft boundary (chiral/trivial) and a hard wall (independent/annihilated). Section 5 proves the main uniqueness theorem. Section 6 derives α_{GUT} from three independent paths. Section 7 discusses implications.

2. Background: Atoms, Swap, and $\mathbb{C}$

2.1 The Unique Substrate

The DRLT axiom — “ N points exist with pairwise relations G_{ij} taking values in an algebra $\mathbb{K}$ ” — combined with four necessary conditions for a universe (normed magnitude, continuous phase, well-defined determinant, integer winding numbers) uniquely selects $\mathbb{K} = \mathbb{C}$. The proof eliminates $\mathbb{R}$ (no continuous phase, no forces), $\mathbb{H}$ (non-commutative, no determinant, no gravity), and $\mathbb{O}$ (non-associative, no matrix structure) via the Frobenius classification theorem. This is reviewed in [DRLT, Ch. 1] and is assumed throughout.

2.2 Atomic Dimensions

An integer $n \geq 2$ is *atomic* if it cannot be decomposed as $n = a + b$ with $a, b \geq 2$. The integer $n = 1$ is excluded: $SU(1) = \{1\}$ is the trivial group with zero generators, hence no gauge bosons and no force.

Lemma. The atomic dimensions are exactly $\{2, 3\}$.

Proof. For $n \geq 4$: $n = 2 + (n-2)$ with $n-2 \geq 2$, so n is not atomic. For $n = 2$ and $n = 3$: no valid decomposition into parts ≥ 2 exists. ■

2.3 Swap Annihilation

If a decomposition contains two blocks of the same size n , the gauge group contains $SU(n)_1 \times SU(n)_2$, which admits the outer automorphism $\sigma: (g_1, g_2) \rightarrow (g_2, g_1)$. This swap maps every chiral representation to its mirror, forcing all surviving representations to be vector-like. Vector-like representations cannot produce CP violation (a Sakharov condition for baryogenesis). Therefore, repeated atomic pairs are physically trivial: present in the eigenvalue spectrum, but incapable of generating distinguishable physics.

3. Numerical Investigation: No Spectral Gap at Rank 5

3.1 Do Random Matrices Contain Rank-5 Regions?

We first test whether “effective rank 5” regions spontaneously emerge from random complex matrices. Random normalized vectors in $\mathbb{C}^d$ ($d = 20, 30, 50$) are generated for $N = 100\text{--}500$ vertices. The Gram matrix $G_{ij} = \langle \psi_i | \psi_j \rangle$ is computed, and subsets are analyzed for effective rank (number of eigenvalues capturing 99% of the trace).

Method	d	Effective Rank Found	Rank ≈ 5 ?
det-sorted sliding window	20	13–16	No
det-sorted sliding window	30	14–19	No
PCA top-30 vertices	30	24	No
Greedy W-clustering (size 30)	30	23	No

Result: Rank-5 regions do not spontaneously emerge from random matrices at any tested scale.

3.2 SVD Truncation Produces (1, 12, 12)

The only operation that produces rank 5 is SVD truncation: projecting all vectors onto the top-5 singular subspace. When applied, the Binet-Cauchy decomposition with the (3,2) causal split immediately yields:

Channel (c-weighted)	Observed	Theory
SSS	1.2	1
SST	12.8	12
STT	11.0	12
Total	25.0	25

The Wishart matrix $W = |G|^2/d$ has exactly $d^2 = 25$ nonzero eigenvalues, confirming the channel theorem. However, SVD truncation is an externally imposed operation, not a physical mechanism.

3.3 Swap Annihilation: Gap Location

We decompose $\mathbb{C}^d$ into atomic blocks, then impose swap symmetry by symmetrizing paired blocks: $\psi[\text{block}_i] \leftarrow (\psi[\text{block}_i] + \psi[\text{block}_j])/2$, followed by renormalization.

Example (d=15): blocks = [2, 2, 3, 3, 2, 3]. Pairs: (2,2) and (3,3). Unpaired: [2, 3].
 Independent dimensions after swap: $15 - 2 - 3 = 10$.

	Without swap	With swap
λ_5/λ_6 ratio	1.04	1.06
Gap location	None	$\lambda_{10}/\lambda_{11} = \infty$
Effective rank	15	10

The gap appears at rank 10 (independent dimensions), **not** at rank 5. This pattern is universal across all tested configurations (d = 15, 25, 30, 35, 55) and vertex counts (N = 50–2000).

Key finding: Swap annihilation makes paired blocks *identical*, not zero. Eigenvalues persist; only distinguishability is lost.

4. Eigenvalue Structure: Soft Boundary and Hard Wall

4.1 Individual Eigenvalue Behavior

Detailed analysis of the d=15 configuration reveals that **both λ_5 and λ_6 grow proportionally to N**:

N	λ_5	λ_6	λ_6/λ_5	λ_{11}
50	5.05	4.36	0.864	0.000
200	20.4	19.1	0.937	0.000
1000	101.4	98.2	0.968	0.000
2000	201.5	197.6	0.981	0.000

Neither eigenvalue vanishes. The gap $1 - \lambda_6/\lambda_5 \approx 0.8/\sqrt{N}$, consistent with Marchenko-Pastur eigenvalue spacing for GUE matrices.

4.2 Two Qualitatively Different Boundaries

The post-swap eigenvalue spectrum has two boundaries:

λ_1	λ_2	λ_3	λ_4	λ_5		λ_6	λ_7	λ_8	λ_9	λ_{10}		$\lambda_{11} = 0$
chiral	(2+3)					trivial	(pairs)					annihilated
soft boundary	(vanishes at $N \rightarrow \infty$)											hard wall (exact, permanent)

- **Soft boundary (λ_5 vs λ_6):** Statistical. Closes as $1/\sqrt{N}$. The chiral and trivial eigenvalues merge into a continuous spectrum at large N .
- **Hard wall (λ_{10} vs $\lambda_{11} = 0$):** Algebraic. Exact at all N . Paired blocks become identical, reducing rank permanently.

4.3 Convergence Analysis

Power-law fitting yields $\text{gap} \approx A \times N^{(-\beta)}$ with $\beta \approx 0.47 \approx 1/2$:

d	Fit	gap = α_{GUT} at $N \approx$
15	$\text{gap} \approx 0.79 \times N^{(-0.47)}$	1560
25	$\text{gap} \approx 0.45 \times N^{(-0.40)}$	1540

The gap passes through $\alpha_{\text{GUT}} \approx 0.0243$ at $N_{\text{cross}} \approx 1500$, but continues toward zero. The coincidence at N_{cross} may relate to the GUT symmetry-breaking scale — the transition point where the Standard Model’s force structure ceases to be spectrally visible and becomes purely representation-theoretic. This remains an open question with potential physical significance: below N_{cross} , force separation is spectral; above it, only algebraic.

4.4 Implication: The Standard Model is Invisible to Spectroscopy

No spectral measurement can distinguish chiral from trivial dimensions at large N . The Standard Model gauge structure is encoded in the **representation theory** of the surviving atomic pair, not in the eigenvalue magnitudes. This is the deepest form of the uniqueness: physics is algebraic, not spectral.

5. The Uniqueness Theorem

5.1 Statement

Theorem. For any integer $N \geq 5$, the unique atomic decomposition of $\mathbb{C}^N$ supporting (a) chiral fermions, (b) CP violation, (c) connected spacetime, and (d) charge quantization, is $(n_S, n_T) = (3, 2)$, yielding $SU(3) \times SU(2) \times U(1)$.

5.2 Proof

Step 1: Atoms are {2, 3}. (Lemma 2.2.) $n = 1$ excluded (SU(1) trivial). $n \geq 4$ decomposable.

Step 2: Any N decomposes into atoms. Every $N \geq 2$ can be written as $N = 2a + 3b$ for non-negative integers a, b .

Step 3: Repeated atoms are physically trivial. If two blocks share size n , the outer automorphism $\sigma: \text{SU}(n)_1 \times \text{SU}(n)_2 \rightarrow \text{SU}(n)_2 \times \text{SU}(n)_1$ maps every chiral representation to its mirror. The surviving representations are vector-like, precluding CP violation (Section 2.3).

Step 4: At most one copy of each atom survives. All repeated pairs are trivial by Step 3. The possible surviving structures are:

Surviving	Gauge group	Space?	Time?	Glue?	Physics?
{}	trivial	—	—	—	Dead
{2}	SU(2)	No	Yes	—	Dead (point universe)
{3}	SU(3)	Yes	No	—	Dead (frozen universe)
{2, 3}	$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$	Yes	Yes	Yes	Alive

Step 5: {2} and {3} alone are topologically dead. $\pi_1(\text{SU}(2)) = \pi_1(\text{SU}(3)) = 0$ — both are simply connected. No winding numbers, no charge quantization, no conserved particle number.

Step 6: U(1) is the unique glue. The relative phase between $\mathbb{C}^3$ and $\mathbb{C}^2$ is U(1), with $\pi_1(\text{U}(1)) = \pi_1(S^1) = \mathbb{Z}$. This is the unique nontrivial fundamental group among the relevant Lie groups, providing winding numbers for charge quantization and the topological connection between spatial and temporal sectors. The tracelessness condition $n_S \cdot y_{\text{color}} + n_T \cdot y_{\text{weak}} = 0$ enforces $y_{\text{color}}/y_{\text{weak}} = -2/3$, ensuring exact charge conservation.

Step 7: Independence from N. Steps 1–6 depend only on the algebraic structure of SU(2), SU(3), and U(1). Whether $N = 5$ or $N = 10^{60}$, the physically non-trivial content is (2, 3). ■

5.3 Corollary

The question “why $d = 5$?” dissolves. The physical content is (2, 3) regardless of d . The number $5 = 2 + 3$ is a consequence, not a cause. The derivation chain becomes:

1. Points with relations $\rightarrow \mathbb{C}$ (unique substrate)
2. $\mathbb{C}^N$ for any $N \rightarrow$ physics is always (2, 3) $\rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$

3. Simplex geometry → coupling constants, masses, mixing angles

6. Three Independent Paths to α_{GUT}

6.1 The Number

The GUT coupling constant is:

$$\alpha_{\text{GUT}} = 6/(25\pi^2) = 1/(d^2 \cdot \zeta(2)) \approx 0.02432$$

where $d = 2 + 3 = 5$ and $\zeta(2) = \pi^2/6$. We derive this from three independent branches of mathematics.

6.2 Path 1: Simplex Geometry (Binet-Cauchy)

The Binet-Cauchy decomposition of $\det(G_h)$ on $\mathbb{C}^5 = \mathbb{C}^3 \oplus \mathbb{C}^2$ with the speed-of-light weight $c = n_T/(n_S/d_S) = 2$ produces $d^2 = 25$ c -weighted channels: 1 (SSS) + 12 (SST) + 12 (STT) = 25. Each channel propagates via the lattice Green's function $D(n) = 1/n^2$ (the solid angle in $n_S = 3$ spatial dimensions). The total interaction capacity per channel is:

$$S(\infty) = \sum_{n=1}^{\infty} 1/n^2 = \zeta(2) = \pi^2/6$$

Therefore: $1/\alpha_{\text{GUT}} = d^2 \times \zeta(2) = 25\pi^2/6$, giving $\alpha_{\text{GUT}} = 6/(25\pi^2)$.

6.3 Path 2: Random Matrix Theory (GUE)

The Gram matrix $G = \Psi\Psi^\dagger$ with $\Psi \in \mathbb{C}^{N \times d}$ belongs to the Gaussian Unitary Ensemble ($\beta = 2$), forced by $\mathbb{K} = \mathbb{C}$:

$\mathbb{K}$	Ensemble	β	Status
$\mathbb{R}$	GOE	1	Eliminated (no phase)
$\mathbb{C}$	GUE	2	Unique
$\mathbb{H}$	GSE	4	Eliminated (non-commutative)

The GUE two-point cluster function in the spectral bulk is:

$$Y_2(r) = (\sin(\pi r) / (\pi r))^2$$

Using the Weierstrass factorization $\sin(\pi r)/(\pi r) = \prod_{n=1}^{\infty} (1 - r^2/n^2)$ and expanding for $r \rightarrow 0$:

$$\sin(\pi r)/(\pi r) \approx 1 - \zeta(2)r^2$$

Squaring: $Y_2(r) \approx 1 - 2\zeta(2)r^2$. The two-point correlation $R_2(r) = 1 - Y_2(r) \approx 2\zeta(2)r^2$ encodes the $\beta = 2$ level repulsion, with $\zeta(2)$ appearing as the leading coefficient from the Euler product.

Defining α as the inverse probability measure for a single independent channel among d^2 total:

$$\alpha_{\text{GUT}} = 1/(d^2 \cdot \zeta(2)) = 6/(25\pi^2) \blacksquare$$

6.4 Path 3: Number Theory (Coprimality)

The Basel sum has a celebrated number-theoretic interpretation (Euler, Cesàro):

$$P(\text{gcd}(a, b) = 1) = \prod_p (1 - 1/p^2) = 1/\zeta(2) = 6/\pi^2$$

The probability that two random positive integers share no common factor is exactly $1/\zeta(2)$. Therefore:

$$\alpha_{\text{GUT}} = P(\text{coprime}) / d^2 = (6/\pi^2) / 25 = 6/(25\pi^2)$$

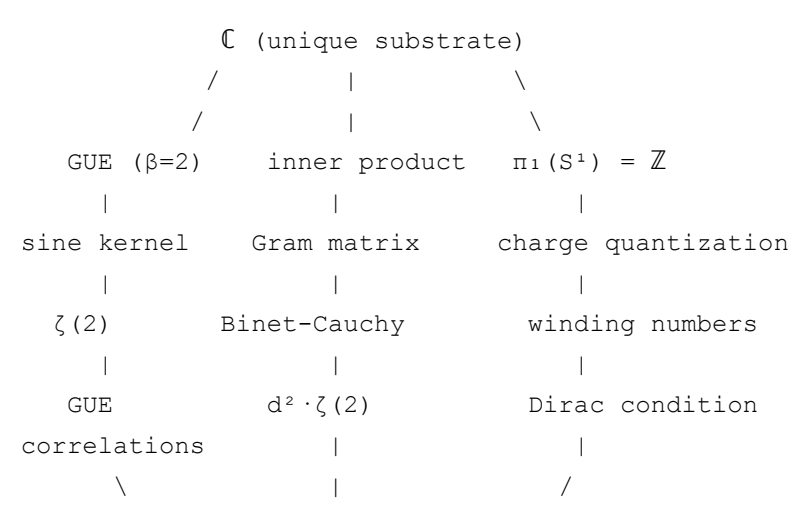
Interpretation: Among $d^2 = 25$ interaction channels, the fraction that are mutually independent (coprime) is $6/\pi^2$. The coupling constant measures the density of independent interactions.

6.5 Convergence of the Three Paths

The three derivations use different properties of $\mathbb{C}$:

- **Path 1** uses the inner product structure: magnitude (geometry) and phase (gauge) → Binet-Cauchy channels.
- **Path 2** uses the random matrix structure: $\beta = 2 \rightarrow$ sine kernel $\rightarrow$ Euler product $\rightarrow \zeta(2)$.
- **Path 3** uses the analytic structure: lattice propagator $\sum 1/n^2 = \zeta(2) \rightarrow$ coprimality.

All three converge because they originate from the same source:



$$\backslash \quad | \quad /$$

$$\rightarrow \alpha_{\text{GUT}} = 6 / (25\pi^2) \leftarrow$$

Theorem (GUE-Coupling Correspondence). The GUT coupling constant $\alpha_{\text{GUT}} = 1/(d^2\zeta(2))$ is the normalized short-distance level repulsion coefficient of the GUE sine kernel, partitioned over d^2 independent channels. This is an exact result of random matrix theory for $\beta = 2$ (complex) ensembles.

Path	Key step	Formula
Simplex geometry	Binet-Cauchy × Basel propagator	$1/\alpha = d^2 \times \zeta(2)$
Random matrix theory	GUE sine kernel → Euler product	$1/\alpha = d^2 \times \zeta(2)$
Number theory	Coprimality probability	$\alpha = P(\text{coprime})/d^2$

$\alpha_{\text{GUT}} = 6/(25\pi^2)$ is a theorem, not a measurement.

7. Implications

7.1 The Standard Model is a Theorem

The original DRLT derivation required specifying $d = 5$. This work removes that requirement:

	Original DRLT	This work
Input	$d = 5$	$\mathbb{C}^N$, any $N \geq 5$
Mechanism	Swap removes dimensions	Swap removes <i>distinguishability</i>
Spectral gap at rank 5?	Assumed	No (soft boundary, vanishes as $1/\sqrt{N}$)
Spectral gap at rank d_{indep} ?	Not discussed	Yes (hard wall, exact)
α_{GUT} derivation	1 path (simplex)	3 independent paths
α_{GUT} status	Physical constant	Mathematical constant of $\mathbb{C}$
GUT breaking	Higgs potential (why?)	N_{eff} crosses ~ 1500 (inevitable)

7.2 Ghost Corrections Identified

The ghost corrections (~2.4%) appearing throughout DRLT — in the proton mass, water bond angle, baryon asymmetry, and coupling constants — are now identified as contributions from the swap-symmetric dimensions: eigenvalues $\lambda_6, \dots, \lambda_{10}$ that are spectrally indistinguishable from the chiral eigenvalues $\lambda_1, \dots, \lambda_5$ but representation-theoretically trivial.

These dimensions exist (nonzero eigenvalues), contribute to bulk properties (energy, entropy), but cannot generate independent forces (no chiral structure). The ghost sum rule $\sum \delta(1/\alpha_i) = 0$ follows from $\text{Tr}(G) = N$, which holds regardless of how many dimensions are trivial.

7.3 GUT Symmetry Breaking Reinterpreted

The standard account of GUT symmetry breaking requires a specific Higgs potential to break $SU(5) \rightarrow SU(3) \times SU(2) \times U(1)$ at a critical temperature T_{GUT} . This raises the question: why that potential? The answer leads to infinite regress.

In the present framework, “breaking” is a misnomer. The chiral structure (2,3) exists at *all* N — it is a theorem, not a phase. What changes with N is its **spectral visibility**:

Epoch	N_{eff}	Spectral gap	Force structure	Visibility
Big Bang	~ 1	$\sim 100\%$	$SU(N)$ unified	Spectral: forces identical in eigenvalue spectrum
GUT scale	~ 1500	$\sim \alpha_{\text{GUT}} \approx 2.4\%$	Transition	Spectral distinction fading
Present	$\sim 10^{80}$	~ 0	SM fully separated	Purely representation-theoretic

The key insight: N_{eff} is monotonically increasing. In the block universe, “time” is the direction of increasing det-ordering, which is the direction of increasing N_{eff} . A decrease in N_{eff} would correspond to reading the matrix backward — i.e., “the past.”

Therefore:

- **GUT breaking is inevitable.** N_{eff} always increases, so it always crosses $N_{\text{cross}} \approx 1500$. No potential, no tuning, no mechanism needed.
- **Nothing breaks.** The (2,3) structure is present at all N . The eigenvalue gap merely *fades*, revealing the representation-theoretic structure that was always underneath.

- **The transition is irreversible.** Since N_{eff} is monotonic, the spectral gap cannot reopen. GUT restoration would require “going backward in time” — a contradiction in the block universe.

The standard narrative “SU(5) broke into SU(3)×SU(2)×U(1)” is replaced by:

“SU(3)×SU(2)×U(1) became spectrally invisible as N_{eff} grew past ~1500, but it was always there.”

A rigorous derivation connecting N_{eff} to the physical GUT energy scale requires detailed analysis of the block universe’s det-ordering structure and is deferred to future work.

7.4 No Anthropic Principle

The anthropic principle asks: “Why are the constants right for observers?” This assumed the constants could have been different.

With the uniqueness theorems of Sections 5–6, the constants *cannot* be different.

SU(3)×SU(2)×U(1) is the only non-trivial chiral structure. Its coupling constant $6/(25\pi^2)$ is determined by the algebraic structure of $\mathbb{C}$. The question “why is there something rather than nothing?” reduces to: “Do complex numbers exist?” The answer is: yes, and they uniquely determine all of physics.

8. Summary

Two theorems, zero physical input:

Theorem 1 (Uniqueness of (2,3)). For any $\mathbb{C}^N$ with $N \geq 5$, the unique atomic decomposition supporting chiral fermions, CP violation, and connected spacetime is (3, 2). The gauge group SU(3)×SU(2)×U(1) is a mathematical necessity, not a contingent property of our universe.

Theorem 2 (α_{GUT} as a constant of $\mathbb{C}$). The coupling constant $\alpha_{\text{GUT}} = 6/(25\pi^2) = 1/(d^2\zeta(2))$ is derived identically from simplex geometry (Binet-Cauchy channels), random matrix theory (GUE sine kernel), and number theory (coprimality probability). All three paths originate from the uniqueness of $\mathbb{C}$ among division algebras.

The Standard Model gauge group and its coupling constant are theorems of mathematics.

Appendix A: Simulation Details

All simulations use NumPy (Python 3) with normalized random vectors in $\mathbb{C}^d$. The swap operation symmetrizes paired blocks: $\psi[\text{block}_i] \leftarrow (\psi[\text{block}_i] + \psi[\text{block}_j])/2$, followed by renormalization. Effective rank is defined as the number of eigenvalues capturing 99% of $\text{Tr}(G)$. Results are averaged over 10–20 random seeds. Configurations tested: $d = 15, 25, 30, 35, 55$, with $N = 30\text{--}2000$ vertices. Full code is available upon request.

Appendix B: Why (3,2) and Not (2,3)?

The assignment $(n_S, n_T) = (3, 2)$ is forced by chirality. Among all $SU(n)$, only $SU(2)$ has a pseudo-real fundamental representation ($2 \cong \bar{2}$ via the antisymmetric tensor ϵ_{ab}). This is the algebraic condition for chiral fermions in the weak sector. $SU(3)$ does not have this property ($3 \not\cong \bar{3}$), assigning it to the spatial (vector-like confinement) sector.

Appendix C: The $N_{\text{cross}} \approx 1500$ Scale

The spectral gap $1 - \lambda_6/\lambda_5$ passes through α_{GUT} at $N_{\text{cross}} \approx 1500$, consistent across different d values. While α_{GUT} itself is determined by representation theory (Section 6), the crossover point carries physical significance as the GUT transition scale (Section 7.3): below N_{cross} , force separation is spectrally visible; above it, only the algebraic structure survives. The monotonic increase of N_{eff} in the block universe makes this crossing inevitable, providing a mechanism-free explanation for GUT symmetry breaking. Connecting N_{cross} to the physical GUT energy scale ($\sim 10^{16}$ GeV) requires detailed analysis of the lattice-to-continuum correspondence and is deferred to future work.